

Exercise set 7

Textbook: Exercises 8.4*, 8.5, 11.3**, 11.6**

*) In 8.4 you may use that the MLE estimator is $\hat{\lambda} = n / \sum_{i=1}^n X_i$, where $X_1, X_2, \dots, X_n$ denote the times between failures.

**) In both cases, check the output you get by calculating the effective sample size, and carry out the diagnostic tests discussed in the lecture notes. For 11.3, you may e.g. use the MALA.

Problem 1:

In this problem, we consider a simple special case of a generalized linear model, where the observations y_i , $i = 1, \dots, n$ are modelled as

$$y_i \sim \text{Exponential}(\exp(\alpha + \beta x_i)), \quad i = 1, \dots, n.$$

where x_i , $i = 1, \dots, n$ is a fixed covariate (e.g. y_i is the price of an appartement, and x_i is size (measured in square-meters) of the appartement). Notice in particular that we use the scale parameterization of the exponential distribution, so $E(y_i) = \exp(\alpha + \beta x_i)$. The parameters α and β may take any values, and we let the parameter vector $\boldsymbol{\theta}$ be defined as $\boldsymbol{\theta} = (\alpha, \beta)$

In this problem, we are going to perform a Bayesian analysis of a data set under this model. We assume the priors $\alpha \sim N(0, 10^2)$ and $\beta \sim N(0, 10^2)$.

The following code simulates a data set, and then defines the function `lp` which gives the log-posterior distribution (up to a normalizing constant)

```
# simulate data
set.seed(123)
n <- 100
x <- runif(n, min=-1)
y <- rexp(n, rate=1.0/exp(0.4+0.5*x))

# log-target distribution (theta=(alpha,beta))
lp <- function(theta){
  alpha <- theta[1]
  beta <- theta[2]
  log.like <- sum(dexp(y, rate=1.0/exp(alpha+beta*x), log=TRUE))
  log.prior <- dnorm(alpha, sd=10.0, log=TRUE) + dnorm(beta, sd=10.0, log=TRUE)
  return(log.like+log.prior)
}
```

Please copy this code into the start of your R-script.

- a) Using the random walk Metropolis Hastings algorithm obtained in the previous exercise set, obtain samples from the posterior distribution $p(\boldsymbol{\theta}|\mathbf{y})$ (the function `lp` above gives the value of $\log g$ where g is proportional to this posterior distribution). Be sure to tune your sampler appropriately.
- b) Before proceeding, check whether you should remove burn-in samples. Check that you have at least 1000 effective samples post-burn-in for both α and β . Check for lack of convergence using the Geweke test.
- c) Suppose you wish to perform a prediction, say y^* for $x = 0.5$. By transforming the MCMC output above, obtain samples from $\exp(\alpha_t + \beta_t \cdot 0.5)$. The distribution you obtain is the distribution of $E(y^*|x = 0.5, \mathbf{y})$ while taking remaining uncertainty (after seeing data) in the parameters into account. We refer to this distribution as the predictive distribution of $E(y^*|x, \mathbf{y})$.
Find the mean and standard deviation of this distribution.
- d) For each MCMC sample t , also simulate new "data" $y_t^* \sim \text{Exponential}(\exp(\alpha_t + \beta_t \cdot 0.5))$. The distribution you obtain is the distribution of a new observation y^* obtained at $x = 0.5$, while taking remaining uncertainty (after seeing data) in the parameters into account. We refer to this distribution as the predictive distribution of $y^*|x = 0.5, \mathbf{y}$.
Find mean and standard deviation of this distribution. Find the probability of obtaining an observation > 5.0 (conditional on $\mathbf{y}$ and for $x = 0.5$).
- e) For $x = -1, -0.95, -0.9, \dots, 0.9, 0.95, 1$ make plots of the median and the 5% and 95% quantiles of the predictive distribution of $E(y^*|x, \mathbf{y})$. (The predictive distributions are obtained in the same way as in c), but for different values of x). An example of how this may look like is given in Figure 1.
- f) Do the same as in e), but for $y^*|x, \mathbf{y}$ this time. An example of output is given in Figure 2. Hint, to get smooth curves as in Figure 2, use the same set of $\text{Exponential}(1)$ random variates to generate new observations for each value of x .

Problem 2:

Consider a univariate Metropolis Hastings algorithm with Normal random walk proposal mechanism, i.e.

$$q(\theta^*|\theta_{t-1}) = \mathcal{N}(\theta^*|\theta_{t-1}, \sigma^2)$$

Show that the general MH accept probability reduces to the RWMH accept probability in this case.

Now consider a uniform random walk proposal, i.e. $\theta^*|\theta_{t-1} \sim \text{Uniform}(\theta_{t-1} - w, \theta_{t-1} + w)$ for some scale parameter w . Does also this MH accept probability reduce to a RWMH accept probability?

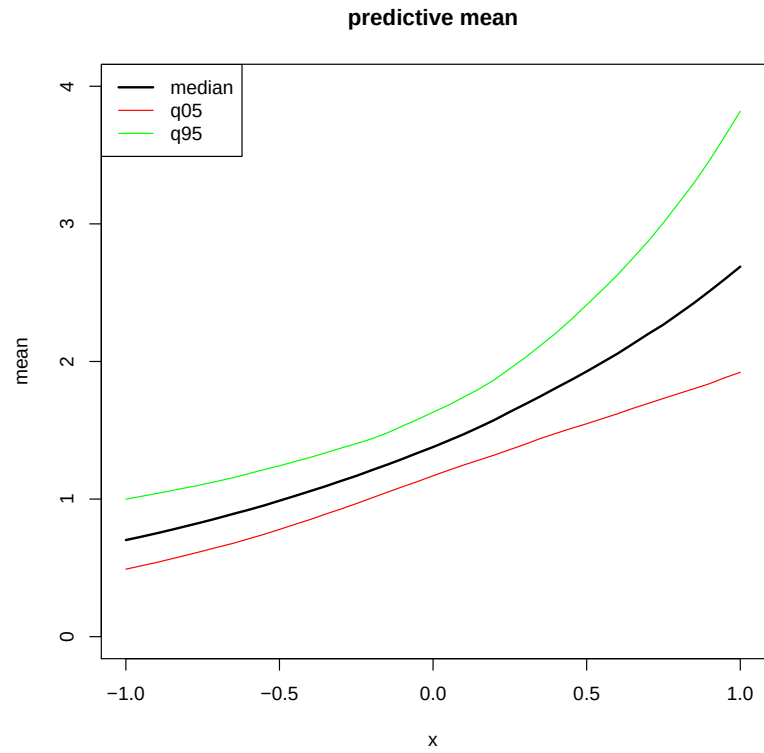


Figure 1: predictive distribution of $E(y^*|x, \mathbf{y})$ for different x

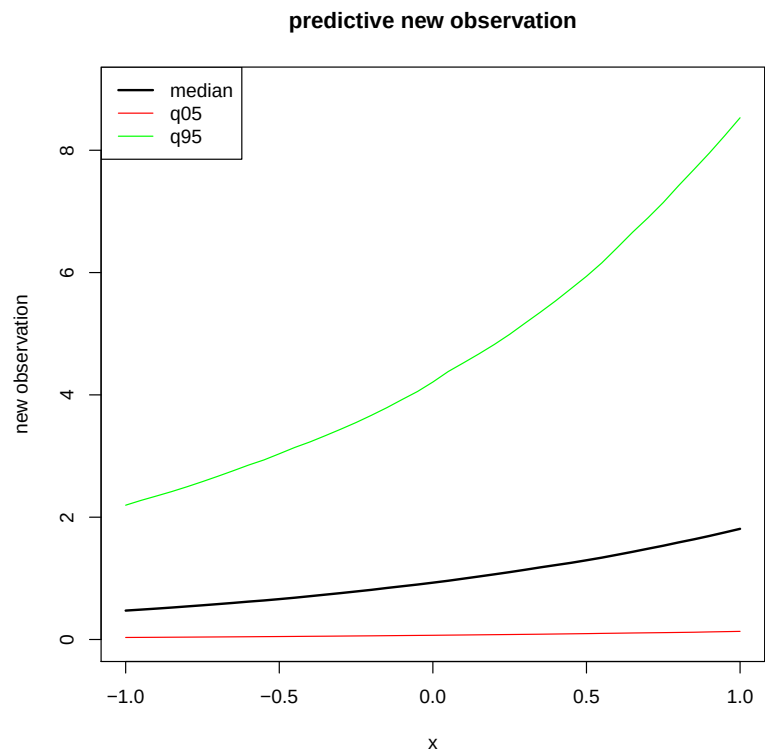


Figure 2: predictive distribution of $y^*|x, \mathbf{y}$ for different x